

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I HARI VISHNU N / 192524319 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

- 1) Construct three address code for the following
 If $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then
 $Z = X + y * z$
 else
 $z = z + 1$

STEP-1: Condition Identification
 $(a < b) \text{ and } ((c < d) \text{ or } (a > d))$

Condition:

$a < b$
 $c < d$
 $a > d$

STEP-2 Table creation

$L_1, L_2, L_3, L_4, L_5, L_6$

STEP-3 TAC for condition

$L_1:$

If $a < b$ goto L_2
 goto L_5

$L_2:$

If $c < d$ goto L_4
 goto L_3

$L_3:$

If $a > d$ goto L_4
 goto L_5

If $a < b$ is false, the complete AND expression is false, so control goes to the else part. If $a < b$ is true expression is evaluated. If either $c < d$ or $a > d$ is true, control goes to the then part.

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STEP-4: Generate TAC for the THEN part
the statement is $z = x + y * 3$. Multiplication has higher precedence
than addition

$L4 \rightarrow \text{Then}$ $L5 \rightarrow \text{else}$

THEN:

$$z = x + y * 3$$

$L4:$

$$t1 = y * 3$$

$$t2 = x + t1$$

$$z = t2$$

goto $L6$

ELSE:

$$z = z + 1$$

$L5:$

$$t3 = z + 1$$

$$z = t3$$

$L6:$

FINAL TAC:

$L1:$

if $a < b$ goto $L2$

goto $L5$

$L2:$

if $c < d$ goto $L4$

goto $L2$

$L3:$

if $a > d$ goto $L4$

goto $L5$

$L4:$

$$t1 = y * 3$$

$$t2 = x + t1$$

$$z = t2$$

goto $L6:$

2) Translate $(a < b) \text{ and } (c < d) \text{ or } (e < f)$ into three add. statements using back patching.

Given boolean expression

$(a < b) \text{ and } (c < d) \text{ or } (e < f)$

STEP-1: Apply operator precedence AND has higher precedence than OR, so the expression is interpreted as:

$[(a < b) \text{ and } (c < d) \text{ or } (e < f)]$

STEP 2:- Generate Incomplete Jump:

if $a < b$ goto _

goto _

if $c < d$ goto _

goto _

if $e < f$ goto _

goto _

STEP 3: HANDLE AND

$(a < b) \text{ AND } (c < d)$ are true if $a, a < b$ is true control must continue to the second condition.

Backpatch statement 1 to statement 3.

if $a < b$ goto 3

STEP 4: HANDLE OR

If the first AND exp. is false the second OR operand $e < f$ must be evaluated.

Backpatch stat. 2 to stat. 5

Backpatch stat. 4 to stat. 5

STEP 5: Find Backpatch code

1. if $a < b$ goto 3

2. goto 5

3. If $c < d$ goto 7

4. goto 5

5. If $e < f$ goto 7

6. goto 8

7. TRUE

8. FALSE

3. Construct TAC using the foll. production for Boolean expression $(a < b) \text{ or } (c < d) \text{ or } (e < f)$

$B \rightarrow B_1 || B_2$

$B \rightarrow B_1$

$B \rightarrow !B$

$B \rightarrow E_1 \text{ relop } E_2$

$B \rightarrow \text{True}$

$B \rightarrow \text{False}$

$[(a > b) \text{ or } (c < d)] \text{ or } (e < f)$

Because this is an OR expression. If any one of the three relation condition is true, the compare expression is TRUE.

Check the first condition:-

L1:

If $a > b$ goto LTrue

goto L2

Check the second condition:

L2:

If $c < d$ goto LTrue

goto L3

Check the third condition:

L3:

If $e < f$ goto LTrue

Find TAC for Q4

L1:

If $a > b$ goto LTrue

goto L2

goto L3

L3:

If $e < f$ goto LTrue

goto LFalse

LTrue:

goto LAND

LEND

4. Generate the three address code for the following Program Fragment:

while ($a < c$ and $B > d$) do

If $A = 1$ then $C = C + 1$

else while $a \leq d$ do

$A = A + B$.

L1:

$t_1 = C + 1$

$C = t_1$

goto L1

L2:

$t_2 = A + B$

$A = t_2$

L2:

$t_3 = A - B$

If $A = 1$ goto L4

goto L5

14:

$t_1 = c + 1$

$c = t_1$

goto 11

15:

If $A \geq D$ goto 16

goto 11

16:

$t_2 = A + B$

$A = t_2$

goto 15

LEND:

N.D.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps